

Mithilesh Chaurasiya

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My Portfolio

GitHub

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Professional Summary

Machine Learning Engineer with hands-on experience building end-to-end ML systems, supported by 2.8 years of experience in data engineering and ETL workflows at Infosys. Specialized in Computer Vision and MLOps, with proven ability to design scalable pipelines using Airflow, MLflow, and DVC. Recently built a production-grade Smart Traffic Management System for real-time traffic analytics.

Technical Skills

- **Programming:** Python
- **Machine Learning & Deep Learning:** Scikit-learn, PyTorch, CNNs, Tensorflow
- **Computer Vision:** OpenCV, YOLO, Object Detection
- **GenAI / LLM:** LangChain, RAG, LangGraph, Tool Calling, AI Agents, Prompt Engineering
- **MLOps & Deployment:** MLflow, DVC,
- **Data Engineering:** Apache Spark, MongoDB, Snowflake
- **Data Analysis:** Pandas, NumPy, SQL (Oracle)
- **Cloud:** Microsoft Azure
- **Tools & Version Control:** Git, GitHub
- **Web Development:** Flask
- **Infrastructure:** Apache Airflow, Docker, Kubernetes (Minikube), FastAPI, CI/CD (Jenkins, Azure DevOps)

Core Strengths

- **Problem Solving:** Strong analytical approach to optimizing data pipelines and ML workflows.
- **Leadership & Mentorship:** Leading the Machine Learning team at DCC, mentoring students on ML, Computer Vision, and real-world project development.
- **Collaboration:** Experience working in Agile teams with cross-functional stakeholders on data-driven systems.
- **Communication:** Ability to translate business requirements into scalable technical solutions.

Education

Master of Computer Application (MCA) 2024 – Expected 2027
National Institute of Technology, Agartala

Bachelor of Science in Computer Science 2021
Thakur College of Science and Commerce, Mumbai
CGPA: 8.87/10

Higher Secondary Certificate (HSC) 2018
Utkarsha Vidyalaya & Junior College, Mumbai
Percentage: 64%

Secondary School Certificate (SSC) 2016
Jai Deep Vidya Mandir High School, Mumbai
Percentage: 75%

Professional Experience

Infosys Ltd. June 2021 – January 2024
Senior System Associate (Data Engineering & ETL Testing)

- Validated and monitored production-grade ETL pipelines handling large-scale structured data across Oracle, MongoDB, and Snowflake, ensuring high data integrity for downstream ML and analytics systems.
- Built Python-based automation frameworks for data validation and API testing, reducing manual QA effort by 30%+ and improving overall pipeline reliability and efficiency.
- Optimized ETL workflows to generate ML-ready datasets, improving data availability and reducing pre-processing time for analytics and machine learning use cases.
- Performed end-to-end validation of REST APIs and model endpoints, ensuring correctness and consistency of model outputs in integrated production systems.
- Collaborated with cross-functional teams to implement ML experiment tracking (MLflow) and dataset versioning (DVC), improving reproducibility, traceability, and consistency of model development workflows.
- Worked in Agile/Scrum environments, contributing to sprint planning, testing strategies, and client-facing deliverables in a cross-functional team setting.

Projects

1. Smart Traffic Management System (End-to-End CV + MLOps Project)

YOLO (Ultralytics), OpenCV, Airflow, MLflow, DVC, MongoDB, FastAPI, Streamlit

- Built a production-grade computer vision + MLOps system for automated traffic analytics.
- Implemented real-time vehicle detection and multi-object tracking for persistent ID assignment.
- Designed traffic intelligence modules for vehicle count, density, and flow estimation.
- Orchestrated complete ML lifecycle (data → training → inference) using Apache Airflow DAGs.
- Integrated DVC for large-scale dataset versioning (22GB IDD dataset) ensuring full reproducibility.
- Implemented MLflow-based experiment tracking with dataset hash → model version traceability.
- Added training deduplication logic to automatically skip duplicate experiments, reducing GPU cost.
- Built Streamlit dashboard for real-time traffic monitoring and analytics visualization.
- Designed modular, production-ready architecture following real-world ML system design principles.

2. DesignKaro — Interactive Distributed Systems Design Workbench (in progress)

React/Next.js, React Flow, Redis, PostgreSQL, LLM integration

- Built an interactive workbench for practicing system design: API gateways, load balancing, caching, and queues,

with traffic simulation and failure injection.

- Implemented capacity planning using Little's Law with SLA/p99 latency analysis.
- Added a rule engine for design feedback plus an LLM-assisted Socratic interviewer.

3. Stock Market ETL Pipeline & Dashboard

Python, Airflow, MongoDB, Streamlit

- Built a complete ETL pipeline to fetch and clean stock market data, storing it in MongoDB.
- Automated daily workflows using Apache Airflow.
- Developed an interactive Streamlit dashboard showing stock price trends, SMA, and volume.
- Optimized MongoDB queries using indexing and caching for faster performance.

4. Real-Time Gender Detection

Python, TensorFlow, OpenCV, CNN

- Developed a CNN-based system to classify gender from facial images with over 90% accuracy.
- Implemented real-time face detection using OpenCV.
- Fine-tuned hyperparameters to optimize model performance.

Leadership & Extracurricular Activities

Machine Learning Team Lead — Developers and Coders Club (DCC)

National Institute of Technology, Agartala

- Leading the Machine Learning team responsible for mentoring junior students in ML, AI, and Data Science.
- Conducted sessions on Machine Learning fundamentals, Deep Learning, and Computer Vision.
- Guided students in building real-world ML projects and Kaggle-based learning.
- Coordinating with faculty mentors on research-oriented and project-based initiatives.
- Planning technical workshops, roadmap sessions, and ML awareness programs for NIT Agartala students.
- Volunteer, College Fest Committee — Assisted in planning and organizing technical events and cultural activities.

Certifications

- Accenture North America - Data Analytics and Visualization Job Simulation
- Infosys Certified Selenium with Python Automation Tester
- Infosys Certified Software Development Engineer in Test (SDET)
- Infosys Certified Python Programmer
- Infosys Certified Data Management Associate

Hobbies

- Reading
- Photography
- Traveling